Thi Mui Pham

u6045188@utah.edu +1 617 300 9017 https://tm-pham.github.io/
tm-pham4984 Google Scholar

EDUCATION

Ph.D. **University Medical Center Utrecht**, Infectious Disease Epidemiology — Utrecht, The Netherlands, Feb 2017 – Feb 2022
- Advisors: Dr. Mirjam Kretzschmar, Martin Bootsma, Ganna Rozhnova
- **Thesis:** Contagions under control: Mathematical models to inform infectious disease prevention and control

M.Sc. **Utrecht University**, Epidemiology — Utrecht, The Netherlands, Sept 2017 – Feb 2022
- Completed concurrently with Ph.D.
- Dutch final grade: 8.1 (GPA $\approx$ 3.7)
- **Coursework**: Clinical Epidemiology, Molecular Epidemiology, Machine learning: Applications in Medicine

M.Sc. **RWTH Aachen**, Major: Mathematics, Minor: Computer Science — Aachen, Germany, Sept 2013 – June 2016
- German final grade: 1.1 (GPA $\approx$ 3.9)
- **Thesis:** Algebraic Statistics for Ordinal Log-Linear Models
- **Coursework:** Decision Theory and Bayesian Inference, Introduction to Artificial Intelligence, Data Mining Algorithms

B.Sc. **RWTH Aachen**, Major: Mathematics, Minor: Computer Science — Aachen, Germany, Oct 2010 – Mar 2013
- Final German grade: 1.8 (GPA $\approx$ 3.2)
- **Thesis:** Algebraic attacks on stream ciphers

RESEARCH EXPERIENCE

University of Utah, Postdoctoral Research Associate — Salt Lake City, UT, USA, Sept 2025 – present
- Advisor: Dr. Katharine Walter
- Leads development of panel regression models to quantify the impact of ambient air pollution on *Mycobacterium tuberculosis* incidence using Brazil's national disease surveillance data (1.7M+ cases, 2003–23); under review
- Leads development of statistical models to estimate and explain the seasonality patterns of TB in Brazil
- Leads development of models to estimate the size of *M. tuberculosis* transmission bottlenecks to inform genomic transmission inference

Harvard T.H. Chan School of Public Health, Postdoctoral Fellow — Boston, MA, USA, Mar 2022 – Aug 2025
- Advisors: Dr. Marc Lipsitch, Dr. Yonatan Grad; Collaborator: Dr. Matthew Samore
- Led analyses of antimicrobial resistance trends across VA Healthcare System, published in *The Lancet Infectious Diseases* and covered by several media
- Led the development of statistical models linking antibiotic prescribing patterns to resistance emergence; published in *The Lancet Microbe*
- Led genomic epidemiological analyses of trends in incidence and resistance for five major causes of bacteremia in the Calgary Health Region (2006–22); under review
- Led integrated analyses of genomic and epidemiological data for real-time SARS-CoV-2 outbreak response in the NBA

Ministry of Education, Netherlands, Infectious Disease Modeler — Utrecht, The Netherlands, Mar 2021 – Oct 2021

- Advisor: Dr. Patricia Bruijning-Verhagen
- Led development of an agent-based model to evaluate SARS-CoV-2 transmission under school reopening scenarios; findings directly informed national policy decisions

University of Oxford, Visiting Researcher — Oxford, UK, Feb 2020 – Aug 2020

- Advisor: Dr. Ben Cooper
- Contributed to UK SAGE Hospital Onset COVID-19 Working Group: developed models to estimate nosocomial transmission burden; informed hospital infection control
- Led the development of statistical models quantifying hand hygiene intervention effectiveness on respiratory infection spread

RWTH Aachen University, Research Assistant — Aachen, Germany, Oct 2014 – Feb 2016

- Developed and implemented randomization methods for clinical trials of rare diseases (IDeAL project) in R package **randomizeR**

AWARDS AND GRANTS

Best Oral Presentation Award at National Postdoctoral Appreciation Week Research Symposium, University of Utah — Sept 2026

Halloran Scholarship to attend two modules of Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) — July 2026

SLAS Visiting Graduate Researcher Program Grant at University of Oxford — Feb 2020 – July 2020

PROMOS scholarship for research internship in Québec City, Canada — Aug 2016 – Nov 2016

Friedrich Ebert Foundation scholarship for Master's degree — Feb 2015 – Jul 2016

Deutschlandstipendium scholarship for Master's degree — Oct 2014 – Feb 2015

DAAD scholarship for Erasmus exchange year in London, UK — Sept 2013 – Jun 2014

START scholarship for committed students with migration background — Aug 2007 – Jul 2009

TEACHING AND MENTORING

Co-instructor, **Advanced Programming in Epidemiology** course at University of Utah. Topics include: efficient and ethical use of artificial intelligence in research, functional and object-oriented programming, debugging, profiling, C++ integration with R, parallel computing, high-performance computing with Slurm, R package development, Git/GitHub.

Formally **mentored** three undergraduate rotation students in tuberculosis epidemiology, University of Utah (2025–26)

Developed and taught a **short course on integrating genomic data into hospital outbreak analyses**, Annual Workshop to Increase Diversity in Mathematical Modeling and Public Health, Harvard (2025)

Taught **medical statistics courses** for medical and economics students, UMC Utrecht (2017–21)

Taught **statistics courses** for medical and economics students, RWTH Aachen (2013–16)

Led **focus groups** in pathogen genomics (University of Utah) and Bayesian statistics (Harvard)

PUBLICATIONS

Pham TM, Zhang Y, Nevers M, Li H, Khader K, Grad YH, Samore M, Lipsitch M (2026). Antimicrobial selection and multidrug resistance in four major pathogens in the US Veterans Affairs health-care system: a multicentre, retrospective, cohort study. *The Lancet Microbe* **7**(9): 101445. DOI: 10.1016/j.lanmic.2026.101445

Chong CE, **Pham TM**, Carey ME, Wee BA, Taouk ML, Favieres JF, Moore CE, Dyson ZA, Lim C, Brown CL, Williamson D, Opatowski L, Outterson K, Mukiri KM, Sherry NL, Essack SY, Brisse S, Grad YH, Baker KS (2024). Conference report of the 2024 Antimicrobial Resistance Meeting. *npj Antimicrobials and Resistance* **2**(1): 43. DOI: 10.1038/s44259-024-00058-z

Pham TM, Zhang Y, Nevers M, Li H, Khader K, Grad YH, Lipsitch M, Samore M (2024). Trends in infection incidence

and antimicrobial resistance in the US Veterans Affairs Healthcare System: a nationwide retrospective cohort study (2007–22). *The Lancet Infectious Diseases* **24**(12): 1333–1346. DOI: 10.1016/s1473-3099(24)00416-x

Cooper BS, Evans S, Jafari Y, **Pham TM**, Mo Y, Lim C, Pritchard MG, Pople D, Hall V, Stimson J, Eyre DW, Read JM, Donnelly CA, Horby P, Watson C, Funk S, Robotham JV, Knight GM (2023). The burden and dynamics of hospital-acquired SARS-CoV-2 in England. *Nature* **623**(7985): 132–138. DOI: 10.1038/s41586-023-06634-z

Jafari Y, Yin M, Lim C, Pople D, Evans S, Stimson J, **Pham TM**, Read JM, Robotham JV, Cooper BS, Knight GM (2022). Effectiveness of infection prevention and control interventions, excluding personal protective equipment, to prevent nosocomial transmission of SARS-CoV-2: a systematic review and call for action. *Infection Prevention in Practice* **4**(1): 100192. DOI: 10.1016/j.infpip.2021.100192

Pham TM, Büchler AC, Voor in 't holt AF, Severin JA, Bootsma MCJ, Gommers D, Kretzschmar ME, Vos MC (2022). Routes of transmission of VIM-positive *P. aeruginosa* in the adult intensive care unit-analysis of 9 years of surveillance at a university hospital using a mathematical model. *Antimicrobial Resistance & Infection Control* **11**(1): 55. DOI: 10.1186/s13756-022-01095-x

Knight GM, **Pham TM**, Stimson J et al. (2022). The contribution of hospital-acquired infections to the COVID-19 epidemic in England in the first half of 2020. *BMC Infectious Diseases* **22**(1): 556. DOI: 10.1186/s12879-022-07490-4

Mo Y, **Pham TM**, Lim C, Horby P, Stewardson AJ, Harbarth S, Scott GM, Cooper BS (2022). The effect of hand hygiene frequency on reducing acute respiratory infections in the community: a meta-analysis. *Epidemiology and Infection* **150**: e79. DOI: 10.1017/s0950268822000516

Teslya A, Rozhnova G, **Pham TM**, van Wees DA, Nunner H, Godijk NG, Bootsma M, Kretzschmar ME (2022). The importance of sustained compliance with physical distancing during COVID-19 vaccination rollout. *Communications Medicine* **2**(1): 146. DOI: 10.1038/s43856-022-00207-3

Pham TM, Mo Y, Cooper BS (2022). The potential impact of intensified community hand hygiene interventions on respiratory tract infections: a modelling study. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* **478**(2261): 20210746. DOI: 10.1098/rspa.2021.0746

Pham TM, Tahir H, van de Wijgert JHHM, Van der Roest BR, Ellerbroek P, Bonten MJM, Bootsma MCJ, Kretzschmar ME (2021). Interventions to control nosocomial transmission of SARS-CoV-2: a modelling study. *BMC Medicine* **19**(1): 211. DOI: 10.1186/s12916-021-02060-y

Pham TM, Westerhof I, Rozhnova G, Bootsma M, Kretzschmar M, Bruijning-Verhagen P (2021). Tweede rapport modellering SARS-CoV-2 infecties in het voortgezet onderwijs. *UMC Utrecht*.

Teslya A[*], **Pham TM**[*], Godijk NG[*], Kretzschmar ME, Bootsma MCJ, Rozhnova G (2020). Impact of self-imposed prevention measures and short-term government-imposed social distancing on mitigating and delaying a COVID-19 epidemic: A modelling study. *PLOS Medicine* **17**(7): e1003166. DOI: 10.1371/journal.pmed.1003166
* Co-first authors

Pham TM, Kateri M (2019). Inference for Ordinal Log-Linear Models Based on Algebraic Statistics. *Journal of Algebraic Statistics* **10**(1): 30–50. DOI: 10.18409/jas.v10i1.74

Pham TM, Kretzschmar M, Bertrand X, Bootsma M, on behalf of COMBACTE-MAGNET Consortium (2019). Tracking *Pseudomonas aeruginosa* transmissions due to environmental contamination after discharge in ICUs using mathematical models. *PLOS Computational Biology* **15**(8): e1006697. DOI: 10.1371/journal.pcbi.1006697

Under Review and in Preparation

Pham TM, Mendonça T, Zhang Y, Mallia D, Croda J, Cohen T, Andrews J, Réquia WJ, Walter KS (2026). Wildfire-driven air pollution and tuberculosis incidence in Brazil: A nationwide panel regression study, 2003–23. *medRxiv*. DOI: 10.64898/2026.07.01.26356762 *Under review.*

Pham TM et al. (2026). Trends in incidence and resistance for five major causes of bacteremia in a Canadian metropolitan area, 2006–22: a genomic and antibiotic use cohort study. *medRxiv*. DOI: 10.64898/2026.08.27.26361471 *Under review.*

Pham TM et al. (2026). Transmission or Importation? Using SARS-CoV-2 Genomics to Guide Infection Control Protocols in the NBA. *In preparation.*

CONFERENCE PRESENTATIONS

Oral presentation at *NPAW Postdoc Research Symposium at UoU*: Impact of wildfire-related fine particulate matter on tuberculosis notifications in Brazil: a nationwide panel study, 2003–23	Sept 2026
Poster presentation at *Society for Epidemiologic Research*: Impact of wildfire-related fine particulate matter on tuberculosis notifications in Brazil: a nationwide panel study, 2003–23	Jun 2026
Oral presentation at *Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies (Wellcome Connecting Science)*: Genomic epidemiology of strain and antibiotic resistance dynamics for five major causes of bacteremia in a Canadian metropolitan area, 2006–22	Mar 2024
Rapid-Fire Poster Presentation at *IDWeek*: Trends in infection incidence and antimicrobial resistance in the US Veterans Affairs Healthcare System: a nationwide retrospective cohort study	Oct 2023
Invited oral presentation at *Preparing for the Next Pandemic Workshop, Banff International Research Station*: Controlling COVID-19 in secondary schools	Jun 2022
Oral presentation at *Annual meeting of the Society for Mathematical Biology*: Impact of self-imposed prevention measures and short-term government-imposed social distancing in mitigating and delaying a COVID-19 epidemic: A modelling study	Jun 2020
Oral presentation at *40th Conference of the International Society for Clinical Biostatistics*: Tracking *Pseudomonas aeruginosa* transmissions in ICUs using mathematical models	Jul 2019

CORE COMPETENCIES

Modeling: Agent-based models, compartmental models, MCMC/Bayesian inference, multi-level regression, panel regression, case-crossover designs

Genomics: Integration of pathogen genomic and epidemiological data, phylodynamic and phylogenetic outbreak analyses (BEAST, phybreak, TransPhylo, outbreakeR)

Programming: R (primary), C++, Python, MATLAB; SQL; Linux/bash; Slurm/HPC; Git; Nextflow

Languages: German (native), English (fluent), Vietnamese and Dutch (conversational), French (basic)

MEDIA COVERAGE

Kahn K. In-Hospital Infections, Antimicrobial Resistance Declined Over 13 Years. *MedPage Today*. Aug 15, 2024.

Emerson L. New Study Warns About Increased Antibiotic Resistance Against Third-Generation Broad-Spectrum Antibiotics in VA Facilities. *U.S. Medicine*. Oct 14, 2024.

Dall C. Study finds significant pre-COVID declines in antibiotic-resistant infections at VA hospitals. *CIDRAP*. Aug 14, 2024.

SERVICE AND OUTREACH

Speaker, Pint of Science Boston (2025) — Presented on mathematical modeling in infectious diseases at a public science festival that brings researchers into pubs and cafés to share their work with general audiences.

Academic Committee Member, 13th Annual Workshop to Increase Diversity in Mathematical Modeling and Public Health, Harvard T.H. Chan School of Public Health (2024–25) — Developed curriculum, led workshop sessions, and participated in candidate selection

Organizing Committee Member, ID Epi Spring Seminar, Harvard T.H. Chan School of Public Health (2023)

Ph.D. Council Member, Graduate School of Life Sciences, Utrecht (2018–22) — Co-authored programme reports, organized annual Ph.D. Day

Curriculum Volunteer, Kiron Open Higher Education (2015–17) — Developed computer science curriculum for refugee students

Finalist, Pitch competition Breaking Science, Utrecht (2019) — Presented on using mathematical models to understand hospital transmission of bacteria at an annual science communication contest for early career researchers.